Vivek Venkatraman Hegde

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EDUCATION

Dr. Ambedkar Institute of Technology **2023–2027**
Bachelor of Engineering, Information Science — CGPA: 8.90 *Bengaluru, Karnataka, India*

TECHNICAL SKILLS

Languages: Python, SQL (Fundamentals)
Frameworks/Libraries: Pandas, NumPy, Matplotlib (Basics)
Data & BI Tools: MS Excel (Basics), Power BI (Basic Dashboards)
Developer Tools: VS Code, GitHub, Git
Concepts: Data Cleaning, Exploratory Data Analysis, Data Visualization, DBMS Fundamentals

COURSEWORK

• DBMS • Data Structures • edx-CS50 Python course *(In Progress)*

PROJECTS

Retail Demand Forecasting & Inventory Analytics (2026) | Python, SQL, Power BI

- Built a Bronze/Silver/Gold ETL pipeline (DuckDB, Parquet) on M5 retail sales data; LightGBM weekly forecasts achieved 36.0% WAPE vs 41.6% for the naive baseline.
- Performed ABC/XYZ product segmentation and reorder-point scenario analysis; built a 3-page Power BI dashboard for sales, forecast and product insights.

GitHub: github.com/vivekhegde-IS/retail-demand-optimization

Multimodal Sensor Fusion for Robust Vehicle Collision Detection | Ongoing

- Developing a real-time vehicle collision detection system using CARLA, camera, IMU/GPS, and RFID sensor fusion.
- Optimizing the perception pipeline for low-latency edge deployment.

AI Research Paper Analysis Platform (2026) | Team Project (4 members)

- Contributed to the development of an AI-powered platform for analyzing and comparing research papers.
- Implemented document processing and AI-based summarization features using Gemini and Grok APIs

GitHub: github.com/Pai05/airesearchagent

CERTIFICATIONS

- **Python (Basic) Certification — HackerRank** *Jun 2026*
- **Google Solution Challenge 2026: Build with AI — Hack2Skill** *Jul 2026*
- **HackerRank Orchestrate — Rank 581 of 1,983 (AI Agent Build)** *Aug 2026*
- **Build4U Karnataka Hackathon — RAG-based Return-Policy Agent** *Sep 2026*

LANGUAGES

English (Fluent), Hindi (Conversational), Kannada (Native)